

## CANDO, ND, USA

WMO: 720858

Lat: 48.480N

Lon: 99.236W

Elev: 1477

StdP: 13.93

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 00285

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -21.9      | -17.6 | -27.4                         | 1.3 | -21.3 | -23.9 | 1.5 | -17.4 | 31.9                  | 13.3 | 28.7 | 9.1  | 7.6                   | 290  | 0.707 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.4                   | 86.0            | 69.6 | 82.4 | 68.4 | 81.1 | 67.3 | 74.4                | 80.7 | 72.1 | 78.9 | 69.9 | 77.1 | 12.2                 | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.5                            | 127.2 | 77.1 | 70.1 | 117.0 | 75.3 | 67.7 | 107.7 | 73.9 | 38.8          | 80.8 | 36.7 | 77.4 | 34.9 | 76.6 | 84.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 28.2          | 25.1 | 22.4 | DB                         | -27.4 | 93.2               | 4.0 | 3.7                                                | -30.3 | 95.9       | -32.6 | 98.1       | -34.8 | 100.2      | -37.7 | 102.9 | (4) |
| (5)               |      |      | WB                         | N/A   | N/A                | N/A | N/A                                                | N/A   | N/A        | N/A   | N/A        | N/A   | N/A        | N/A   | N/A   | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 38.2   | 7.7   | 6.3   | 22.9  | 36.8  | 53.0  | 63.2  | 67.5  | 65.3  | 56.5  | 41.0  | 24.8  | 11.8  |
|                                                                                  | DBStd       | 24.32  | 14.09 | 12.78 | 14.41 | 10.47 | 8.96  | 5.64  | 5.05  | 5.66  | 8.12  | 8.58  | 11.53 | 13.03 |
|                                                                                  | HDD50       | 6122   | 1312  | 1223  | 843   | 406   | 70    | 1     | 0     | 0     | 29    | 300   | 755   | 1183  |
|                                                                                  | HDD65       | 10027  | 1777  | 1643  | 1306  | 846   | 385   | 98    | 33    | 68    | 275   | 743   | 1205  | 1648  |
|                                                                                  | CDD50       | 1834   | 0     | 0     | 3     | 9     | 163   | 396   | 542   | 474   | 224   | 22    | 1     | 0     |
|                                                                                  | CDD65       | 261    | 0     | 0     | 0     | 0     | 12    | 43    | 110   | 77    | 19    | 0     | 0     | 0     |
|                                                                                  | CDH74       | 2977   | 0     | 0     | 0     | 3     | 261   | 450   | 1045  | 929   | 283   | 6     | 0     | 0     |
|                                                                                  | CDH80       | 731    | 0     | 0     | 0     | 0     | 71    | 74    | 235   | 278   | 73    | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 10.7   | 11.9  | 11.2  | 11.0  | 12.1  | 11.9  | 10.4  | 8.5   | 8.1   | 9.9   | 11.5  | 10.9  | 11.0  |
| Precipitation                                                                    | PrecAvg     | 19.4   | 0.4   | 0.5   | 0.8   | 1.2   | 2.8   | 3.7   | 3.3   | 2.5   | 1.9   | 1.5   | 0.8   | 0.6   |
|                                                                                  | PrecMax     | 24.8   | 1.3   | 2.0   | 2.1   | 2.6   | 6.8   | 7.9   | 9.3   | 6.8   | 4.6   | 4.6   | 3.1   | 1.2   |
|                                                                                  | PrecMin     | 12.2   | 0.0   | 0.0   | 0.2   | 0.1   | 0.9   | 1.2   | 1.1   | 1.0   | 0.3   | 0.0   | 0.0   | 0.2   |
|                                                                                  | PrecStd     | 3.7    | 0.3   | 0.4   | 0.6   | 0.7   | 1.5   | 1.7   | 1.9   | 1.3   | 1.1   | 1.2   | 0.9   | 0.3   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 38.8  | 37.1  | 63.4  | 72.5  | 87.7  | 85.5  | 88.5  | 92.5  | 86.3  | 73.0  | 58.8  | 43.0  |
|                                                                                  |             | MCWB   | 35.3  | 34.9  | 53.4  | 52.8  | 63.3  | 69.7  | 73.1  | 71.8  | 67.7  | 57.6  | 49.6  | 36.1  |
|                                                                                  | 2%          | DB     | 35.9  | 33.5  | 55.0  | 64.3  | 81.3  | 81.7  | 84.3  | 86.2  | 81.5  | 66.0  | 49.7  | 36.4  |
|                                                                                  |             | MCWB   | 33.7  | 32.5  | 48.1  | 49.9  | 61.4  | 66.5  | 72.3  | 69.7  | 64.6  | 52.5  | 44.3  | 33.5  |
|                                                                                  | 5%          | DB     | 32.2  | 30.2  | 47.7  | 59.2  | 75.3  | 79.3  | 82.1  | 82.1  | 76.8  | 60.7  | 44.5  | 32.2  |
|                                                                                  |             | MCWB   | 31.1  | 29.4  | 43.6  | 47.6  | 56.5  | 65.3  | 70.7  | 68.1  | 62.1  | 51.2  | 40.0  | 31.4  |
|                                                                                  | 10%         | DB     | 28.1  | 27.0  | 40.8  | 54.4  | 71.8  | 76.5  | 80.8  | 79.4  | 72.4  | 55.3  | 39.0  | 28.3  |
|                                                                                  |             | MCWB   | 27.4  | 25.8  | 37.7  | 44.9  | 55.9  | 64.0  | 69.3  | 66.7  | 60.9  | 48.1  | 36.5  | 27.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 35.9  | 35.2  | 55.6  | 54.9  | 67.0  | 74.7  | 79.5  | 75.5  | 73.0  | 60.7  | 50.3  | 37.9  |
|                                                                                  |             | MCDB   | 38.5  | 37.3  | 60.1  | 68.2  | 77.3  | 79.7  | 84.8  | 83.6  | 79.2  | 66.8  | 57.0  | 41.2  |
|                                                                                  | 2%          | WB     | 33.5  | 32.3  | 48.3  | 51.8  | 64.5  | 71.4  | 75.3  | 72.7  | 69.0  | 55.4  | 44.7  | 33.9  |
|                                                                                  |             | MCDB   | 35.3  | 33.0  | 54.4  | 61.9  | 73.9  | 76.8  | 80.8  | 81.7  | 75.2  | 61.0  | 49.4  | 36.3  |
|                                                                                  | 5%          | WB     | 31.0  | 28.9  | 42.7  | 48.6  | 61.4  | 68.5  | 73.0  | 70.2  | 65.6  | 51.6  | 40.1  | 31.5  |
|                                                                                  |             | MCDB   | 31.9  | 29.8  | 47.7  | 58.6  | 71.3  | 74.9  | 78.8  | 79.0  | 74.3  | 59.6  | 43.3  | 32.6  |
|                                                                                  | 10%         | WB     | 27.7  | 25.5  | 37.8  | 45.8  | 58.2  | 66.2  | 71.0  | 68.2  | 62.6  | 48.8  | 36.9  | 28.1  |
|                                                                                  |             | MCDB   | 28.0  | 26.9  | 40.5  | 52.8  | 66.4  | 73.4  | 78.0  | 76.6  | 69.7  | 56.1  | 39.0  | 28.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 18.1  | 19.0  | 18.6  | 19.9  | 23.0  | 20.2  | 21.4  | 24.6  | 23.6  | 19.7  | 16.4  | 15.8  |
|                                                                                  |             | MCDBR  | 20.8  | 19.5  | 25.3  | 31.0  | 31.8  | 24.9  | 24.1  | 28.7  | 30.3  | 29.8  | 23.0  | 19.3  |
|                                                                                  |             | MCWBR  | 19.1  | 18.0  | 17.5  | 17.3  | 14.7  | 13.1  | 13.0  | 14.8  | 15.5  | 16.6  | 16.8  | 16.4  |
|                                                                                  | 5% WB       | MCDBR  | 20.8  | 18.7  | 24.3  | 29.2  | 25.5  | 20.4  | 21.7  | 25.8  | 27.0  | 25.9  | 22.6  | 19.0  |
| MCWBR                                                                            |             | 19.3   | 17.5  | 17.3  | 17.1  | 13.5  | 13.9  | 13.9  | 14.5  | 15.5  | 15.9  | 16.7  | 16.4  |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.276  | 0.285 | 0.295 | 0.355 | 0.368 | 0.373 | 0.380 | 0.381 | 0.347 | 0.295 | 0.266 | 0.246 |       |
|                                                                                  | taud        | 2.277  | 2.283 | 2.429 | 2.327 | 2.350 | 2.385 | 2.371 | 2.353 | 2.428 | 2.552 | 2.493 | 2.428 |       |
|                                                                                  | Ebn at Noon | 250    | 277   | 295   | 284   | 283   | 282   | 278   | 272   | 272   | 274   | 259   | 254   |       |
|                                                                                  | Edh at Noon | 23     | 29    | 30    | 37    | 38    | 37    | 37    | 36    | 30    | 22    | 19    | 17    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 507    | 874   | 1257  | 1577  | 1790  | 1931  | 2048  | 1689  | 1244  | 778   | 458   | 351   |       |
|                                                                                  | RadStd      | 36     | 80    | 107   | 127   | 114   | 131   | 104   | 110   | 83    | 96    | 63    | 42    |       |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (21 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page